Kartikey Singh

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Summary

Computer Science Engineering graduate specializing in Python, AI/ML, and backend development. Built and deployed production-grade NLP systems using BERT, GPT-2, and Google Gemini API. Experienced in Flask-based web applications, REST APIs, and database-driven architectures.

Education

Dr. A.P.J. Abdul Kalam Technical University — Lucknow, UP
Bachelor of Technology in Computer Science Engineering | CGPA: 7 — *August 2022 – May 2026*

Projects

SatyaX | *AI-Powered Fact Verification & Misinformation Analysis Platform* — June 2026

- **Tech:** Python, Flask, BERT, GPT-2, Google Gemini API, NLP, Clerk Authentication, SQLite/PostgreSQL
- Built a multi-stage AI verification pipeline to detect fake news and assess credibility of news articles and web content.
- Fine-tuned a BERT-based classification model achieving 93% accuracy for fake news detection across multiple article categories.
- Integrated Google Gemini API for contextual reasoning, claim analysis, evidence summarization, and truth-score generation.
- Built URL-based content analysis using automated web scraping and preprocessing pipelines to extract and process article text.
- Demonstrated generative AI misuse risks by implementing a GPT-2 misinformation simulation module for educational awareness.
- Designed shareable analysis reports with PDF export and per-user report history management.
- Implemented secure authentication, profile management, and role-based access control using Clerk Authentication.

Smart Scrap | *AI-Based Waste Management Platform* — July 2025

- **Tech:** Python, Machine Learning, Flask, MySQL, REST APIs, HTML/CSS
- Developed a web-based waste management platform with ML-based automated waste classification and a reward-driven recycling system.
- Trained a classification model on structured waste datasets, achieving 94% accuracy across multiple waste categories.
- Designed separate user and admin workflows for waste reporting, tracking, verification, and incentive management.
- Built backend features for managing user records, waste submissions, and classification results using Flask and MySQL.
- Collaborated in a team of [N] across requirement analysis, development, testing, and deployment phases.

Research & Publications

Advanced International Journal for Research — Volume 7, Issue 2
Smart Scrap: An AI-Based Waste Segregation and Reward-Driven Recycling Platform — *March – April 2026*

- Co-authored a research paper on AI-driven waste segregation, integrating ML-based classification with a reward-driven recycling mechanism.
- Contributed to system design, dataset handling, and AI model integration for automated waste categorization.

Skills

Programming Languages: Python, Java
Web & API Development: Flask, REST APIs, API Integration
Databases: MySQL, Database Management
AI/LLM: PyTorch, Transformers, BERT, GPT-2, Natural Language Processing, Machine Learning, Google Gemini API
CS Fundamentals: Data Structures, Algorithms, Object-Oriented Programming
Developer Tools: Git, GitHub, VS Code, Google Colab, Claude Code, Clerk
Productivity & BI: MS Excel, MS PowerPoint, MS Word, Power BI *(Beginner)*

Certificates & Training

Artificial Intelligence and Machine Learning Program: Indian Council for Technical Research & Development (June 2026)
IBM Project-Based Learning: Real-world ML projects with IBM mentorship (July – Aug 2025)
Data Science & Machine Learning: Comprehensive program via Udemy (May 2025)
Foundations of Project Management: Google Professional Certificate via Coursera (March 2025)
Java Programming Fundamentals: Infosys Springboard (July 2024)